Numerical Linear Algebra II: Lecture Notes

Eigenvalues, SVD, PCA, PageRank, Sparse Reorderings, and Krylov Subspace Methods

Jakub Kružík
jakub.kruzik@vsb.cz

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1 Eigenvalues, Eigenvectors, and Matrix Diagonalization

1.1 Definition and Characteristic Polynomial

Definition 1.1 (Eigenvalues and Eigenvectors)**.** Let $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ be a square matrix. A non-zero vector $\boldsymbol{v} \in \mathbb{C}^n$ ($\boldsymbol{v} \neq \boldsymbol{0}$) is called an *eigenvector* of $\boldsymbol{A}$, and $\lambda \in \mathbb{C}$ is its corresponding *eigenvalue*, if

$$\boldsymbol{A}\boldsymbol{v} = \lambda \boldsymbol{v}. \tag{1}$$

Geometrically, multiplying an eigenvector $\boldsymbol{v}$ by the matrix $\boldsymbol{A}$ does not change its direction (line of action); it only scales the vector by the factor λ.

Rewriting (1) as a homogeneous linear system gives

$$(\boldsymbol{A} - \lambda \boldsymbol{I})\boldsymbol{v} = \boldsymbol{0}.$$

Since we require a non-trivial solution $\boldsymbol{v} \neq \boldsymbol{0}$, the matrix $\boldsymbol{A}-\lambda\boldsymbol{I}$ must be singular. Consequently, λ is an eigenvalue of $\boldsymbol{A}$ if and only if it is a root of the *characteristic polynomial*:

$$p(\lambda) = \det(\boldsymbol{A} - \lambda \boldsymbol{I}) = 0. \tag{2}$$

1.2 Trace, Determinant, and Basic Spectral Properties

The eigenvalues of $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ are closely related to its matrix invariants:

- **Trace:** The sum of the diagonal entries equals the sum of the eigenvalues:

$$\operatorname{tr}(\boldsymbol{A}) = \sum_{i=1}^{n} a_{ii} = \sum_{i=1}^{n} \lambda_i.$$

- **Determinant:** The determinant equals the product of the eigenvalues:

$$\det(\boldsymbol{A}) = \prod_{i=1}^{n} \lambda_i.$$

  For a 2×2 matrix, expanding $\det(\boldsymbol{A} - \lambda \boldsymbol{I})$ yields $\lambda^2 - \operatorname{tr}(\boldsymbol{A})\lambda + \det(\boldsymbol{A}) = 0$. When computing eigenvalues by hand, however, writing out the standard determinant $\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = 0$ directly is generally preferable (as it often preserves factored terms such as $(a - \lambda)^2 - b^2$), while $\sum \lambda_i = \operatorname{tr}(\boldsymbol{A})$ and $\prod \lambda_i = \det(\boldsymbol{A})$ serve as a quick verification check.

- **Spectral Shift:** Adding $\alpha \boldsymbol{I}$ ($\alpha \in \mathbb{R}$) shifts all eigenvalues by α while preserving the eigenvectors:
$$(\boldsymbol{A} + \alpha \boldsymbol{I})\boldsymbol{v} = \boldsymbol{A}\boldsymbol{v} + \alpha \boldsymbol{I}\boldsymbol{v} = \lambda \boldsymbol{v} + \alpha \boldsymbol{v} = (\lambda + \alpha)\boldsymbol{v}.$$
- **Inverse Matrix:** If $\boldsymbol{A}$ is invertible (i.e., $\lambda_i \neq 0$ for all i), multiplying $\boldsymbol{A}\boldsymbol{v} = \lambda \boldsymbol{v}$ by $\boldsymbol{A}^{-1}$ and dividing by λ yields
$$\boldsymbol{v} = \lambda \boldsymbol{A}^{-1}\boldsymbol{v} \quad \Longrightarrow \quad \boldsymbol{A}^{-1}\boldsymbol{v} = \frac{1}{\lambda}\boldsymbol{v}.$$
Thus, $\boldsymbol{A}^{-1}$ has the same eigenvectors as $\boldsymbol{A}$ with eigenvalues $1/\lambda_i$.
- **Matrix Powers:** Multiplying $\boldsymbol{A}\boldsymbol{v} = \lambda \boldsymbol{v}$ repeatedly by $\boldsymbol{A}$ gives
$$\boldsymbol{A}^2\boldsymbol{v} = \boldsymbol{A}(\boldsymbol{A}\boldsymbol{v}) = \lambda \boldsymbol{A}\boldsymbol{v} = \lambda^2 \boldsymbol{v}, \quad \text{and in general,} \quad \boldsymbol{A}^k\boldsymbol{v} = \lambda^k \boldsymbol{v}.$$
- **Complex Conjugate Pairs for Real Matrices:** For *any* real matrix $\boldsymbol{A} \in \mathbb{R}^{n\times n}$, the characteristic polynomial $p(\lambda) = \det(\boldsymbol{A} - \lambda \boldsymbol{I})$ has real coefficients, so any non-real eigenvalues must occur in **complex conjugate pairs** λ and $\overline{\lambda}$. Equivalently, taking the complex conjugate of $\boldsymbol{A}\boldsymbol{v} = \lambda \boldsymbol{v}$ and using $\overline{\boldsymbol{A}} = \boldsymbol{A}$ gives
$$\overline{\boldsymbol{A}\boldsymbol{v}} = \overline{\lambda \boldsymbol{v}} \quad \Longrightarrow \quad \boldsymbol{A}\overline{\boldsymbol{v}} = \overline{\lambda}\,\overline{\boldsymbol{v}},$$
showing that if $(\lambda, \boldsymbol{v})$ is an eigenpair of a real matrix $\boldsymbol{A}$, then $(\overline{\lambda}, \overline{\boldsymbol{v}})$ is also an eigenpair.
- **Symmetry and Skew-Symmetry:**
  - If $\boldsymbol{A}$ is real and **symmetric** ($\boldsymbol{A} = \boldsymbol{A}^T \in \mathbb{R}^{n\times n}$), then all eigenvalues are real ($\lambda_i \in \mathbb{R}$) and there exists a full set of real, mutually orthogonal eigenvectors ($\boldsymbol{v}_i \in \mathbb{R}^n$).
  - If $\boldsymbol{Q}$ is real and **skew-symmetric** (anti-symmetric, $\boldsymbol{Q}^T = -\boldsymbol{Q}$), its eigenvalues are purely imaginary (or zero) and, since $\boldsymbol{Q}$ is real, occur in complex conjugate pairs $\pm i\omega$.

Example 1.1 (Permutation/Reflection Matrix)**.** Consider the symmetric matrix
$$\boldsymbol{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$
Computing the characteristic polynomial via $\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = 0$:
$$\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = \begin{vmatrix} -\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 - 1 = (\lambda - 1)(\lambda + 1) = 0 \quad \Longrightarrow \quad \lambda_1 = 1, \quad \lambda_2 = -1,$$
with corresponding eigenvectors:
$$\boldsymbol{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \lambda_1 = 1; \qquad \boldsymbol{v}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \quad \lambda_2 = -1.$$
As a check, $\lambda_1 + \lambda_2 = 1 + (-1) = 0 = \operatorname{tr}(\boldsymbol{A})$ and $\lambda_1\lambda_2 = -1 = \det(\boldsymbol{A})$.

Example 1.2 (Shifted Symmetric Matrix)**.** Consider the matrix obtained by adding $3\boldsymbol{I}$ to the previous example:
$$\boldsymbol{A} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}.$$
The characteristic polynomial is
$$\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = \begin{vmatrix} 3 - \lambda & 1 \\ 1 & 3 - \lambda \end{vmatrix} = (3 - \lambda)^2 - 1 = \lambda^2 - \underbrace{6}_{\operatorname{tr}(\boldsymbol{A})}\lambda + \underbrace{8}_{\det(\boldsymbol{A})} = (\lambda - 4)(\lambda - 2) = 0.$$
Thus $\lambda_1 = 4$ and $\lambda_2 = 2$ (which are indeed $1 + 3$ and $-1 + 3$ from the shift property $\boldsymbol{A} + 3\boldsymbol{I}$). Solving $(\boldsymbol{A} - \lambda_i \boldsymbol{I})\boldsymbol{v}_i = \boldsymbol{0}$:

- For $\lambda_1 = 4$: $\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \mathbf{0} \implies \boldsymbol{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.
- For $\lambda_2 = 2$: $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \mathbf{0} \implies \boldsymbol{v}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$.

Example 1.3 (Rotation / Skew-Symmetric Matrix)**.** Consider the 90° counter-clockwise rotation matrix in $\mathbb{R}^2$, which maps $\boldsymbol{x} = (1, 0)^T$ to $\boldsymbol{Q}\boldsymbol{x} = (0, 1)^T$:

$$\boldsymbol{Q} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Since $\boldsymbol{Q}$ rotates every non-zero real vector by 90°, no real vector stays on its own line. Indeed, solving the characteristic equation gives

$$\det(\boldsymbol{Q} - \lambda \boldsymbol{I}) = \begin{vmatrix} -\lambda & -1 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 + 1 = 0 \quad \implies \quad \lambda_1 = i, \quad \lambda_2 = -i,$$

which are complex conjugate purely imaginary numbers (satisfying $\lambda_1 + \lambda_2 = 0 = \operatorname{tr}(\boldsymbol{Q})$ and $\lambda_1 \lambda_2 = 1 = \det(\boldsymbol{Q})$).

1.3 Matrix Diagonalization

Suppose $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ has n linearly independent eigenvectors $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n$ with corresponding eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$. Placing the eigenvectors as columns into a matrix $\boldsymbol{Q} = (\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n)$ gives

$$\boldsymbol{A}\boldsymbol{Q} = \boldsymbol{A}(\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n) = (\lambda_1 \boldsymbol{v}_1, \lambda_2 \boldsymbol{v}_2, \ldots, \lambda_n \boldsymbol{v}_n) = (\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n) \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} = \boldsymbol{Q}\boldsymbol{\Lambda},$$

where $\boldsymbol{\Lambda} = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. Since the columns of $\boldsymbol{Q}$ are linearly independent, $\boldsymbol{Q}$ is invertible, yielding the *eigenvalue decomposition (diagonalization)*:

$$\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^{-1}. \tag{3}$$

If $\boldsymbol{A}$ is symmetric ($\boldsymbol{A} = \boldsymbol{A}^T$), the eigenvectors can be chosen orthonormal, so $\boldsymbol{Q}$ is orthogonal ($\boldsymbol{Q}^{-1} = \boldsymbol{Q}^T$) and $\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^T$.

Proposition 1.1 (Sufficient Condition for Diagonalizability)**.** *If all eigenvalues of $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ are distinct ($\lambda_i \neq \lambda_j$ for all $i \neq j$), then the corresponding eigenvectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$ are linearly independent, and $\boldsymbol{A}$ is diagonalizable.*

Remark 1.1 (Defective Matrices). Not every matrix is diagonalizable. For example, the upper triangular matrix

$$\boldsymbol{A} = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$$

has a repeated eigenvalue $\lambda_1 = \lambda_2 = 3$, but solving $(\boldsymbol{A} - 3\boldsymbol{I})\boldsymbol{v} = \mathbf{0}$ yields only a single independent eigenvector direction $\boldsymbol{v} = (1, 0)^T$. Such a matrix lacks n linearly independent eigenvectors and cannot be diagonalized.

Remark 1.2 (Repeated Eigenvalues and the Identity Matrix). Having repeated eigenvalues does not prevent a matrix from being diagonalizable; distinct eigenvalues are a sufficient condition, not a necessary one. For instance, the identity matrix

$$\boldsymbol{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

has the repeated eigenvalue $\lambda_1 = \lambda_2 = 1$, yet every non-zero vector $\boldsymbol{v} \in \mathbb{R}^2$ is an eigenvector ($\boldsymbol{I}\boldsymbol{v} = 1 \cdot \boldsymbol{v}$). Consequently, $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ is diagonalizable by *any* invertible matrix $\boldsymbol{Q}$, since $\boldsymbol{Q}\boldsymbol{I}\boldsymbol{Q}^{-1} = \boldsymbol{I}$.

Remark 1.3 (Algebraic vs. Geometric Multiplicity). The difference between the two examples above is captured by the two notions of multiplicity for an eigenvalue λ:

- **Algebraic multiplicity** $m_a(\lambda)$: the multiplicity of λ as a root of the characteristic polynomial $p(\lambda) = \det(\boldsymbol{A} - \lambda \boldsymbol{I})$.
- **Geometric multiplicity** $m_g(\lambda)$: the dimension of the eigenspace $\mathcal{N}(\boldsymbol{A} - \lambda \boldsymbol{I})$, i.e., the number of linearly independent eigenvectors associated with λ.

For every eigenvalue λ, these multiplicities satisfy

$$1 \leq m_g(\lambda) \leq m_a(\lambda).$$

A matrix $\boldsymbol{A} \in \mathbb{R}^{n \times n}$ is diagonalizable (over $\mathbb{C}$) **if and only if** $m_g(\lambda) = m_a(\lambda)$ for every eigenvalue λ:

- For $\boldsymbol{A} = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$, $\lambda = 3$ has $m_a(3) = 2$ but $m_g(3) = 1 < m_a(3)$, so $\boldsymbol{A}$ is *defective* (not diagonalizable).
- For $\boldsymbol{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $\lambda = 1$ has $m_a(1) = 2$ and $m_g(1) = 2 = m_a(1)$, so $\boldsymbol{I}$ is diagonalizable.

Using diagonalization (3), computing matrix powers becomes trivial:

$$\boldsymbol{A}^2 = (\boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^{-1})(\boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^{-1}) = \boldsymbol{Q}\boldsymbol{\Lambda}^2\boldsymbol{Q}^{-1}, \qquad \boldsymbol{A}^k = \boldsymbol{Q}\boldsymbol{\Lambda}^k\boldsymbol{Q}^{-1},$$

where $\boldsymbol{\Lambda}^k = \operatorname{diag}(\lambda_1^k, \ldots, \lambda_n^k)$.

1.4 Applications to Difference Equations and Markov Chains

Consider a linear discrete-time dynamical system (difference equation):

$$\boldsymbol{u}_{k+1} = \boldsymbol{A}\boldsymbol{u}_k \quad \Longrightarrow \quad \boldsymbol{u}_1 = \boldsymbol{A}\boldsymbol{u}_0, \quad \boldsymbol{u}_2 = \boldsymbol{A}\boldsymbol{u}_1 = \boldsymbol{A}^2\boldsymbol{u}_0, \quad \ldots, \quad \boldsymbol{u}_k = \boldsymbol{A}^k\boldsymbol{u}_0.$$

If $\boldsymbol{A}$ has linearly independent eigenvectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$, we can expand the initial state $\boldsymbol{u}_0$ in the eigenvector basis:

$$\boldsymbol{u}_0 = c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2 + \cdots + c_n\boldsymbol{v}_n = \boldsymbol{Q}\boldsymbol{c}, \quad \text{where } \boldsymbol{c} = (c_1, c_2, \ldots, c_n)^T = \boldsymbol{Q}^{-1}\boldsymbol{u}_0.$$

Applying $\boldsymbol{A}^k$ then gives an explicit closed-form formula for any step k:

$$\boldsymbol{u}_k = \boldsymbol{A}^k\boldsymbol{u}_0 = c_1\lambda_1^k\boldsymbol{v}_1 + c_2\lambda_2^k\boldsymbol{v}_2 + \cdots + c_n\lambda_n^k\boldsymbol{v}_n = \boldsymbol{Q}\boldsymbol{\Lambda}^k\boldsymbol{c}. \tag{4}$$

Stability of Difference Equations. From (4), the long-term behavior of $\boldsymbol{u}_k$ as $k \to \infty$ is governed entirely by the magnitudes $|\lambda_i|$ of the eigenvalues of $\boldsymbol{A}$:

- **Stable (Asymptotically Stable):** $|\lambda_i| < 1$ for all $i = 1, \ldots, n$. Every term $\lambda_i^k \to 0$ as $k \to \infty$, so $\boldsymbol{u}_k \to \boldsymbol{0}$ for every initial state $\boldsymbol{u}_0$.

- **Neutrally Stable:** $|\lambda_i| \leq 1$ for all i, with at least one eigenvalue satisfying $|\lambda_i| = 1$ (and having linearly independent eigenvectors). Modes with $|\lambda_i| < 1$ decay to zero, while modes with $|\lambda_i| = 1$ neither grow nor decay — in particular, if $\lambda_1 = 1$ and $|\lambda_i| < 1$ for $i \geq 2$, the system converges to a non-zero **steady state** $\boldsymbol{u}_\infty = c_1\boldsymbol{v}_1$ (as in Markov chains below).
- **Unstable:** $|\lambda_i| > 1$ for at least one i. The corresponding mode λ_i^k grows exponentially in magnitude as $k \to \infty$, so $\|\boldsymbol{u}_k\| \to \infty$ (whenever $c_i \neq 0$, as in the Fibonacci sequence below).

Example 1.4 (Unstable Difference Equation: Binet's Formula for the Fibonacci Sequence)**.** Consider the Fibonacci sequence $0, 1, 1, 2, 3, 5, 8, 13, \ldots$ defined by $F_0 = 0$, $F_1 = 1$, and the second-order recurrence

$$F_{k+2} = F_{k+1} + F_k.$$

To write this as a first-order system $\boldsymbol{u}_{k+1} = \boldsymbol{A}\boldsymbol{u}_k$, define the state vector $\boldsymbol{u}_k = \begin{pmatrix} F_{k+1} \\ F_k \end{pmatrix}$ with initial state $\boldsymbol{u}_0 = \begin{pmatrix} F_1 \\ F_0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$:

$$\begin{cases} F_{k+2} = F_{k+1} + F_k \\ F_{k+1} = F_{k+1} \end{cases} \iff \boldsymbol{u}_{k+1} = \begin{pmatrix} F_{k+2} \\ F_{k+1} \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}}_{\boldsymbol{A}} \begin{pmatrix} F_{k+1} \\ F_k \end{pmatrix} = \boldsymbol{A}\boldsymbol{u}_k.$$

1. **Eigenvalues:**

$$\det(\boldsymbol{A} - \lambda\boldsymbol{I}) = \begin{vmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 - \lambda - 1 = 0 \implies \lambda_{1,2} = \frac{1 \pm \sqrt{1+4}}{2} = \frac{1 \pm \sqrt{5}}{2}.$$

   Thus $\lambda_1 = \frac{1+\sqrt{5}}{2} \approx 1.618 > 1$ (the golden ratio, an unstable growing mode) and $\lambda_2 = \frac{1-\sqrt{5}}{2} \approx -0.618$ ($|\lambda_2| < 1$, a decaying mode). Since $\lambda_1 \neq \lambda_2$, $\boldsymbol{A}$ is diagonalizable.

2. **Eigenvectors:** From $(\boldsymbol{A} - \lambda\boldsymbol{I})\boldsymbol{v} = \begin{pmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{pmatrix}\boldsymbol{v} = \boldsymbol{0}$, the second row gives $v_1 - \lambda v_2 = 0$, so we can choose

$$\boldsymbol{v}_1 = \begin{pmatrix} \lambda_1 \\ 1 \end{pmatrix}, \qquad \boldsymbol{v}_2 = \begin{pmatrix} \lambda_2 \\ 1 \end{pmatrix}.$$

3. **Initial Condition Coefficients:** We solve $\boldsymbol{u}_0 = c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$:

$$\left(\begin{array}{cc|c} \lambda_1 & \lambda_2 & 1 \\ 1 & 1 & 0 \end{array}\right) \sim \left(\begin{array}{cc|c} 1 & 1 & 0 \\ \lambda_1 & \lambda_2 & 1 \end{array}\right) \sim \left(\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & \lambda_2 - \lambda_1 & 1 \end{array}\right).$$

   Since $\lambda_2 - \lambda_1 = \frac{1-\sqrt{5}}{2} - \frac{1+\sqrt{5}}{2} = -\sqrt{5}$, we get

$$c_2(-\sqrt{5}) = 1 \implies c_2 = -\frac{1}{\sqrt{5}}, \qquad c_1 = -c_2 = \frac{1}{\sqrt{5}}.$$

4. **Closed-Form Solution:** Substituting into $\boldsymbol{u}_k = \boldsymbol{A}^k\boldsymbol{u}_0 = c_1\lambda_1^k\boldsymbol{v}_1 + c_2\lambda_2^k\boldsymbol{v}_2$:

$$\begin{pmatrix} F_{k+1} \\ F_k \end{pmatrix} = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^k \begin{pmatrix} \lambda_1 \\ 1 \end{pmatrix} - \left(\frac{1-\sqrt{5}}{2}\right)^k \begin{pmatrix} \lambda_2 \\ 1 \end{pmatrix}\right].$$

Reading off the second component yields *Binet's formula*:

$$\boxed{F_k = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^k - \left(\frac{1-\sqrt{5}}{2}\right)^k\right].}$$

For large k (e.g., F_{100}), since $|\lambda_2| \approx 0.618 < 1$, the second term vanishes and $F_k \approx \frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^k$.

Markov Matrices and Steady States. An important class of **neutrally stable** difference equations $\boldsymbol{v}_{t+1} = \boldsymbol{A}\boldsymbol{v}_t$ arises from Markov chains, where $\boldsymbol{A}$ describes state transition probabilities.

Definition 1.2 (Markov / Column-Stochastic Matrix)**.** A square matrix $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ is called a *Markov matrix* (or *column-stochastic matrix*) if:

1. All entries are non-negative: $a_{ij} \geq 0$ for all i, j.
2. Every column sums to 1: $\sum_{i=1}^{n} a_{ij} = 1$ for all $j = 1, \ldots, n$ (equivalently, $\boldsymbol{e}^T\boldsymbol{A} = \boldsymbol{e}^T$ where $\boldsymbol{e} = (1, \ldots, 1)^T$).

Key spectral properties of a Markov matrix $\boldsymbol{A}$:

- **Conservation of total probability/population:** Because $\boldsymbol{e}^T\boldsymbol{A} = \boldsymbol{e}^T$, multiplying any state vector $\boldsymbol{v}_t$ by $\boldsymbol{A}$ preserves the sum of its entries:

$$\boldsymbol{e}^T\boldsymbol{v}_{t+1} = \boldsymbol{e}^T(\boldsymbol{A}\boldsymbol{v}_t) = (\boldsymbol{e}^T\boldsymbol{A})\boldsymbol{v}_t = \boldsymbol{e}^T\boldsymbol{v}_t.$$

- **Dominant eigenvalue $\lambda_1 = 1$ (Neutral Stability):** Since $\boldsymbol{A} - \boldsymbol{I}$ has columns summing to zero, $\det(\boldsymbol{A} - \boldsymbol{I}) = 0$, so $\lambda_1 = 1$ is always an eigenvalue of $\boldsymbol{A}$. Moreover, all other eigenvalues satisfy $|\lambda_i| \leq 1$ (and if all entries of $\boldsymbol{A}$, or of some power $\boldsymbol{A}^k$, are strictly positive, then $|\lambda_i| < 1$ for all $i \geq 2$ by the Perron–Frobenius theorem).
- **Steady state as the $\lambda = 1$ eigenvector:** As $t \to \infty$, all modes with $|\lambda_i| < 1$ decay to zero ($\lambda_i^t \to 0$), so $\boldsymbol{v}_{t+1} = \boldsymbol{A}\boldsymbol{v}_t$ converges to a **steady-state vector** $\boldsymbol{v}_\infty$ satisfying

$$\boldsymbol{A}\boldsymbol{v}_\infty = 1 \cdot \boldsymbol{v}_\infty = \boldsymbol{v}_\infty.$$

  That is, the steady-state distribution is precisely the eigenvector of $\boldsymbol{A}$ corresponding to $\lambda_1 = 1$ (normalized so its entries sum to the total population or 1).

Example 1.5 (Neutrally Stable Difference Equation: Population Migration Steady State[1])**.** Suppose that every year a fraction p of New Yorkers move to California (so $1 - p$ stay in New York), and a fraction q of Californians move to New York (so $1 - q$ stay in California). The population vector $\boldsymbol{v}_t = (\mathrm{NY}_t, \mathrm{CA}_t)^T$ evolves according to the discrete-time Markov system:

$$\boldsymbol{v}_{t+1} = \boldsymbol{A}\boldsymbol{v}_t, \qquad \begin{pmatrix}\text{New York}_{t+1}\\ \text{California}_{t+1}\end{pmatrix} = \begin{pmatrix}1-p & q\\ p & 1-q\end{pmatrix}\begin{pmatrix}\text{New York}_t\\ \text{California}_t\end{pmatrix}. \tag{5}$$

Notice that each column of $\boldsymbol{A}$ is non-negative and sums to 1: $(1-p)+p = 1$ and $q+(1-q) = 1$.

Now take $p = 0.3$ and $q = 0.1$:

$$\boldsymbol{A} = \begin{pmatrix}1-0.3 & 0.1\\ 0.3 & 1-0.1\end{pmatrix} = \begin{pmatrix}0.7 & 0.1\\ 0.3 & 0.9\end{pmatrix}.$$

[1] Adapted from the interactive visual explanation https://setosa.io/ev/eigenvectors-and-eigenvalues/.

- **Eigenvalues:** Solving the characteristic equation $\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = 0$:

$$\begin{aligned}\det(\boldsymbol{A} - \lambda \boldsymbol{I}) &= \begin{vmatrix} 0.7 - \lambda & 0.1 \\ 0.3 & 0.9 - \lambda \end{vmatrix} = (0.7 - \lambda)(0.9 - \lambda) - 0.03 \\ &= \lambda^2 - 1.6\lambda + 0.6 = (\lambda - 1)(\lambda - 0.6) = 0,\end{aligned}$$

  which gives

$$\lambda_1 = 1, \qquad \lambda_2 = 0.6 \quad \text{(in general, } \lambda_2 = 1 - p - q\text{)}.$$

- **Eigenvectors:**
  - For $\lambda_1 = 1$:

$$(\boldsymbol{A} - \boldsymbol{I})\boldsymbol{v}_1 = \begin{pmatrix} -0.3 & 0.1 \\ 0.3 & -0.1 \end{pmatrix} \boldsymbol{v}_1 = \boldsymbol{0} \quad \Longrightarrow \quad \boldsymbol{v}_1 = \begin{pmatrix} 1 \\ 3 \end{pmatrix} \quad \left(\text{or } \boldsymbol{v}_1 = \begin{pmatrix} q \\ p \end{pmatrix} \right).$$

    Normalizing the entries to sum to 1 gives the **steady-state probability vector**:

$$\boldsymbol{v}_\infty = \frac{1}{1+3} \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 0.25 \\ 0.75 \end{pmatrix} \quad \left(\text{in general, } \boldsymbol{v}_\infty = \frac{1}{p+q} \begin{pmatrix} q \\ p \end{pmatrix} \right).$$

  - For $\lambda_2 = 0.6$:

$$(\boldsymbol{A} - 0.6\boldsymbol{I})\boldsymbol{v}_2 = \begin{pmatrix} 0.1 & 0.1 \\ 0.3 & 0.3 \end{pmatrix} \boldsymbol{v}_2 = \boldsymbol{0} \quad \Longrightarrow \quad \boldsymbol{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

- **Convergence to Steady State:** For any initial population distribution $\boldsymbol{v}_0$ whose entries sum to 1 (e.g., $\boldsymbol{v}_0 = (1, 0)^T$), expanding $\boldsymbol{v}_0$ in the eigenvector basis yields

$$\boldsymbol{v}_0 = \begin{pmatrix} 0.25 \\ 0.75 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad \Longrightarrow \quad \boldsymbol{v}_t = \boldsymbol{A}^t \boldsymbol{v}_0 = 1^t \begin{pmatrix} 0.25 \\ 0.75 \end{pmatrix} + c_2 (0.6)^t \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

  As $t \to \infty$, $(0.6)^t \to 0$, so regardless of the initial population split, the system converges to the steady state where 25% of the population lives in New York and 75% lives in California:

$$\boldsymbol{A}\boldsymbol{v}_\infty = \begin{pmatrix} 0.7 & 0.1 \\ 0.3 & 0.9 \end{pmatrix} \begin{pmatrix} 0.25 \\ 0.75 \end{pmatrix} = \begin{pmatrix} 0.175 + 0.075 \\ 0.075 + 0.675 \end{pmatrix} = \begin{pmatrix} 0.25 \\ 0.75 \end{pmatrix} = \boldsymbol{v}_\infty.$$

  At this equilibrium, the fraction of the total population moving from New York to California each year ($p \cdot 0.25 = 0.3 \times 0.25 = 0.075$) exactly balances the fraction moving from California to New York ($q \cdot 0.75 = 0.1 \times 0.75 = 0.075$).

1.5 Applications to Systems of Differential Equations

Consider a system of first-order linear ordinary differential equations (ODEs):

$$\frac{d\boldsymbol{u}}{dt} = \boldsymbol{A}\boldsymbol{u}(t), \qquad \boldsymbol{u}(0) = \boldsymbol{u}_0.$$

If $\boldsymbol{A}$ has eigenvalues λ_i and eigenvectors $\boldsymbol{x}_i$ (or $\boldsymbol{v}_i$), the general solution is a linear combination of exponential modes:

$$\boldsymbol{u}(t) = c_1 e^{\lambda_1 t} \boldsymbol{x}_1 + c_2 e^{\lambda_2 t} \boldsymbol{x}_2 + \cdots + c_n e^{\lambda_n t} \boldsymbol{x}_n. \tag{6}$$

Remark 1.4 (Stability of ODE Systems vs. Difference Equations). Since $\left|e^{\lambda_i t}\right| = e^{\mathrm{Re}(\lambda_i)t}$, the stability threshold for continuous-time ODEs $\frac{d\boldsymbol{u}}{dt} = \boldsymbol{A}\boldsymbol{u}$ is $\mathrm{Re}(\lambda_i) = 0$ (rather than $|\lambda_i| = 1$ for discrete-time difference equations):

- **Stable:** $\mathrm{Re}(\lambda_i) < 0$ for all i ($\boldsymbol{u}(t) \to \boldsymbol{0}$ as $t \to \infty$).
- **Neutrally stable:** $\mathrm{Re}(\lambda_i) \leq 0$ for all i, with at least one non-defective eigenvalue having $\mathrm{Re}(\lambda_i) = 0$.
- **Unstable:** $\mathrm{Re}(\lambda_i) > 0$ for at least one i ($\|\boldsymbol{u}(t)\| \to \infty$ as $t \to \infty$).

Example 1.6 (Continuous-Time Compartment System[2])**.** Consider the coupled ODE system:

$$\begin{cases} \dfrac{du_1}{dt} = -u_1 + 2u_2, \\ \dfrac{du_2}{dt} = \quad u_1 - 2u_2, \end{cases} \qquad \text{with initial condition } \boldsymbol{u}(0) = \begin{pmatrix} u_1(0) \\ u_2(0) \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ at } t = 0.$$

In matrix form, $\frac{d\boldsymbol{u}}{dt} = \boldsymbol{A}\boldsymbol{u}$ with

$$\boldsymbol{A} = \begin{pmatrix} -1 & 2 \\ 1 & -2 \end{pmatrix}.$$

- **Eigenvalues:** Solving the characteristic equation $\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = 0$:

  $$\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = \begin{vmatrix} -1-\lambda & 2 \\ 1 & -2-\lambda \end{vmatrix} = (-1-\lambda)(-2-\lambda) - 2 = \lambda^2 + 3\lambda = \lambda(\lambda+3) = 0,$$

  which gives

  $$\lambda_1 = 0, \qquad \lambda_2 = -3 \quad \text{(neutrally stable: } \lambda_1 = 0, \lambda_2 < 0\text{)}.$$

  (Notice that the columns of $\boldsymbol{A}$ sum to zero, expressing conservation of total mass $u_1(t) + u_2(t) = \text{const}$ and guaranteeing $\lambda_1 = 0$.)

- **Eigenvectors:**
  - For $\lambda_1 = 0$: $\begin{pmatrix} -1 & 2 \\ 1 & -2 \end{pmatrix} \boldsymbol{x}_1 = \boldsymbol{0} \implies \boldsymbol{x}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$.
  - For $\lambda_2 = -3$: $\begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix} \boldsymbol{x}_2 = \boldsymbol{0} \implies \boldsymbol{x}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

- **Initial Condition and Steady State:** At $t = 0$,

  $$\boldsymbol{u}(0) = c_1 \begin{pmatrix} 2 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \implies \quad c_1 = \frac{1}{3}, \quad c_2 = \frac{1}{3}.$$

  Therefore, the solution for any $t \geq 0$ is

  $$\boldsymbol{u}(t) = \frac{1}{3} e^{0 \cdot t} \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \frac{1}{3} e^{-3t} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

  As $t \to \infty$, $e^{-3t} \to 0$, and the system converges to the steady state $\boldsymbol{u}(\infty) = \begin{pmatrix} 2/3 \\ 1/3 \end{pmatrix}$.

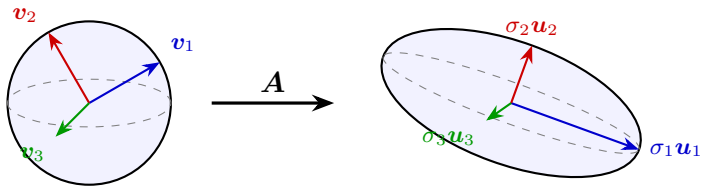
[2] A *compartmental model* (or *compartment system*) divides a system into distinct, well-mixed "compartments" (such as interconnected fluid tanks, organs in pharmacokinetics, or population states in epidemiology) between which mass or individuals flow at specified rates. In this two-compartment example, substance leaves compartment 1 for compartment 2 at rate $1 \cdot u_1$ and flows from compartment 2 back to compartment 1 at rate $2 \cdot u_2$; because the system is closed, every outflow from one compartment is an inflow to the other, so the columns of $\boldsymbol{A}$ sum to zero and the total amount $u_1(t) + u_2(t)$ is conserved.

2 Singular Value Decomposition (SVD)

2.1 Geometric Motivation and Derivation

Whereas eigenvalue decomposition $\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^{-1}$ (or $\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^T$ for symmetric matrices) applies only to square matrices, the *Singular Value Decomposition (SVD)* generalizes orthogonal diagonalization to any rectangular matrix $\boldsymbol{A} \in \mathbb{R}^{m\times n}$.

Geometric Motivation: Consider the action of $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ on the unit sphere $\{\boldsymbol{x} \in \mathbb{R}^n : \|\boldsymbol{x}\|_2 = 1\}$. The image $\{\boldsymbol{A}\boldsymbol{x} : \|\boldsymbol{x}\|_2 = 1\}$ is a hyperellipsoid in $\mathbb{R}^m$:



Unit sphere $\|\boldsymbol{x}\|_2 = 1$ Ellipsoid $\{\boldsymbol{A}\boldsymbol{x} : \|\boldsymbol{x}\|_2 = 1\}$

The orthogonal semi-axes of this ellipsoid have directions given by unit vectors $\boldsymbol{u}_i \in \mathbb{R}^m$ and lengths $\sigma_i \geq 0$, which are the images of orthonormal vectors $\boldsymbol{v}_i \in \mathbb{R}^n$ on the unit sphere:

$$\boldsymbol{A}\boldsymbol{v}_i = \sigma_i \boldsymbol{u}_i.$$

In matrix form, $\boldsymbol{A}\boldsymbol{V} = \boldsymbol{U}\boldsymbol{\Sigma}$, and since $\boldsymbol{V}$ has orthonormal columns ($\boldsymbol{V}^T\boldsymbol{V} = \boldsymbol{I}$), multiplying by $\boldsymbol{V}^T$ on the right gives

$$\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T. \tag{7}$$

Algebraic Derivation: Let $\boldsymbol{A} \in \mathbb{R}^{m\times n}$ with $\operatorname{rank}(\boldsymbol{A}) = r$. The matrix $\boldsymbol{A}^T\boldsymbol{A} \in \mathbb{R}^{n\times n}$ is symmetric positive semi-definite, so it has real non-negative eigenvalues and orthonormal eigenvectors:

$$\boldsymbol{A}^T\boldsymbol{A}\boldsymbol{v}_i = \lambda_i \boldsymbol{v}_i, \quad \|\boldsymbol{v}_i\|_2 = 1, \quad i = 1, \ldots, n,$$

ordered such that $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_r > 0$ and $\lambda_{r+1} = \cdots = \lambda_n = 0$.

1. For $i = 1, \ldots, r$, define the *singular values* $\sigma_i = \sqrt{\lambda_i} > 0$.

2. Multiplying $\boldsymbol{A}^T\boldsymbol{A}\boldsymbol{v}_i = \lambda_i\boldsymbol{v}_i$ on the left by $\boldsymbol{A}$ shows that $\boldsymbol{w}_i = \boldsymbol{A}\boldsymbol{v}_i$ is an eigenvector of $\boldsymbol{A}\boldsymbol{A}^T$ with eigenvalue λ_i:
$$(\boldsymbol{A}\boldsymbol{A}^T)(\boldsymbol{A}\boldsymbol{v}_i) = \lambda_i(\boldsymbol{A}\boldsymbol{v}_i).$$

3. The vectors $\{\boldsymbol{w}_i\}_{i=1}^r$ are mutually orthogonal and have length $\|\boldsymbol{w}_i\|_2 = \sqrt{\boldsymbol{v}_i^T\boldsymbol{A}^T\boldsymbol{A}\boldsymbol{v}_i} = \sqrt{\lambda_i} = \sigma_i$. Normalizing them defines the *left singular vectors*
$$\boldsymbol{u}_i = \frac{\boldsymbol{A}\boldsymbol{v}_i}{\sigma_i}, \quad i = 1, \ldots, r \quad \Longleftrightarrow \quad \boldsymbol{A}\boldsymbol{v}_i = \sigma_i\boldsymbol{u}_i.$$

4. Completing $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_r\}$ to an orthonormal basis $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m\}$ of $\mathbb{R}^m$ (where $\boldsymbol{u}_{r+1}, \ldots, \boldsymbol{u}_m$ span $\mathcal{N}(\boldsymbol{A}\boldsymbol{A}^T) = \mathcal{N}(\boldsymbol{A}^T)$) yields orthogonal matrices $\boldsymbol{U} \in \mathbb{R}^{m\times m}$ and $\boldsymbol{V} \in \mathbb{R}^{n\times n}$ such that
$$\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T, \qquad \boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_r & \boldsymbol{O} \\ \boldsymbol{O} & \boldsymbol{O} \end{pmatrix} \in \mathbb{R}^{m\times n}, \quad \boldsymbol{\Sigma}_r = \operatorname{diag}(\sigma_1, \ldots, \sigma_r).$$

2.2 Key Properties and Eckart–Young–Mirsky Theorem

From $\boldsymbol{A} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$, we immediately obtain:

- **Relation to $\boldsymbol{A}^T\boldsymbol{A}$ and $\boldsymbol{A}\boldsymbol{A}^T$:**

$$\boldsymbol{A}^T\boldsymbol{A} = \boldsymbol{V}\boldsymbol{\Sigma}^T\boldsymbol{U}^T\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T = \boldsymbol{V}(\boldsymbol{\Sigma}^T\boldsymbol{\Sigma})\boldsymbol{V}^T, \qquad \boldsymbol{A}\boldsymbol{A}^T = \boldsymbol{U}(\boldsymbol{\Sigma}\boldsymbol{\Sigma}^T)\boldsymbol{U}^T.$$

  Hence $\sigma_i = \sqrt{\lambda_i(\boldsymbol{A}^T\boldsymbol{A})} = \sqrt{\lambda_i(\boldsymbol{A}\boldsymbol{A}^T)}$ for $i = 1, \ldots, r$.

- **Symmetric Matrices:** If $\boldsymbol{A} = \boldsymbol{A}^T$, then $\sigma_i = |\lambda_i(\boldsymbol{A})|$.

- **Rank-1 Outer Product Expansion:**

$$\boldsymbol{A} = \sum_{i=1}^{r} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^T.$$

- **Spectral Norm and Best Low-Rank Approximation (Eckart–Young–Mirsky Theorem):** The 2-norm of $\boldsymbol{A}$ is $\|\boldsymbol{A}\|_2 = \sigma_1$. For any $k \leq r$, the truncated SVD

$$\boldsymbol{A}_k = \sum_{i=1}^{k} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^T$$

  is the best rank-k approximation to $\boldsymbol{A}$ in the 2-norm, with error equal to the $(k+1)$-st singular value:

$$\|\boldsymbol{A} - \boldsymbol{A}_k\|_2 = \left\| \sum_{i=1}^{r} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^T - \sum_{i=1}^{k} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^T \right\|_2 = \left\| \sum_{i=k+1}^{r} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^T \right\|_2 = \sigma_{k+1}.$$

2.3 Computing the SVD via Golub–Kahan Bidiagonalization

While $\sigma_i = \sqrt{\lambda_i(\boldsymbol{A}^T\boldsymbol{A})}$ defines the singular values theoretically, **explicitly forming $\boldsymbol{A}^T\boldsymbol{A}$ is numerically unstable** because it squares the condition number: $\kappa_2(\boldsymbol{A}^T\boldsymbol{A}) = \kappa_2(\boldsymbol{A})^2$, causing severe loss of accuracy for small singular values in floating-point arithmetic.

Instead, practical SVD algorithms (such as the Golub–Kahan–Reinsch algorithm) proceed in two phases without ever forming $\boldsymbol{A}^T\boldsymbol{A}$:

1. **Phase 1: Reduction to Upper Bidiagonal Form.** Using alternating left and right Householder reflections $\boldsymbol{P}_1, \boldsymbol{P}_2, \ldots$ and $\boldsymbol{Q}_1, \boldsymbol{Q}_2, \ldots$, the matrix $\boldsymbol{A}$ is reduced to an *upper bidiagonal* matrix $\boldsymbol{B}$:

$$\boldsymbol{A} = \boldsymbol{P}\boldsymbol{B}\boldsymbol{Q}^T, \qquad \text{where } \boldsymbol{P}, \boldsymbol{Q} \text{ are orthogonal and } \boldsymbol{B} = \begin{pmatrix} \times & \times & 0 & 0 \\ 0 & \times & \times & 0 \\ 0 & 0 & \times & \times \\ 0 & 0 & 0 & \times \end{pmatrix}.$$

   Specifically, for a 4×4 matrix:

   - Multiply on the left by a Householder reflector $\boldsymbol{P}_1$ to zero out entries below the diagonal in column 1:

$$\boldsymbol{A} = \begin{pmatrix} \times & \times & \times & \times \\ \times & \times & \times & \times \\ \times & \times & \times & \times \\ \times & \times & \times & \times \end{pmatrix} \xrightarrow{\boldsymbol{P}_1^T\boldsymbol{A}} \begin{pmatrix} \times & \times & \times & \times \\ 0 & \times & \times & \times \\ 0 & \times & \times & \times \\ 0 & \times & \times & \times \end{pmatrix}.$$

- Multiply on the right by a Householder reflector $\boldsymbol{Q}_1$ acting on columns $2, \ldots, n$ to zero out entries to the right of the superdiagonal in row 1 (without destroying the zeros in column 1!):

$$\xrightarrow{\boldsymbol{P}_1^T \boldsymbol{A} \boldsymbol{Q}_1} \begin{pmatrix} \times & \times & 0 & 0 \\ 0 & \times & \times & \times \\ 0 & \times & \times & \times \\ 0 & \times & \times & \times \end{pmatrix}.$$

- Repeat for the remaining columns and rows to obtain $\boldsymbol{B} = \boldsymbol{P}^T \boldsymbol{A} \boldsymbol{Q}$.

2. **Phase 2: SVD of the Bidiagonal Matrix.** Note that $\boldsymbol{A}^T \boldsymbol{A} = \boldsymbol{Q} \boldsymbol{B}^T \boldsymbol{P}^T \boldsymbol{P} \boldsymbol{B} \boldsymbol{Q}^T = \boldsymbol{Q} (\boldsymbol{B}^T \boldsymbol{B}) \boldsymbol{Q}^T$, so $\boldsymbol{B}$ has the exact same singular values as $\boldsymbol{A}$. Dedicated iterative algorithms (implicit-shift QR / Golub–Kahan SVD step, or divide-and-conquer) compute the SVD of the bidiagonal matrix $\boldsymbol{B}$:

$$\boldsymbol{B} = \boldsymbol{X} \boldsymbol{\Sigma} \boldsymbol{Y}^T \quad \text{(computed implicitly without forming } \boldsymbol{B}^T \boldsymbol{B}\text{)}.$$

Substituting $\boldsymbol{B} = \boldsymbol{X} \boldsymbol{\Sigma} \boldsymbol{Y}^T$ back into $\boldsymbol{A} = \boldsymbol{P} \boldsymbol{B} \boldsymbol{Q}^T$ yields the full SVD of $\boldsymbol{A}$:

$$\boldsymbol{A} = \boldsymbol{P} (\boldsymbol{X} \boldsymbol{\Sigma} \boldsymbol{Y}^T) \boldsymbol{Q}^T = \underbrace{(\boldsymbol{P} \boldsymbol{X})}_{\boldsymbol{U}} \boldsymbol{\Sigma} \underbrace{(\boldsymbol{Q} \boldsymbol{Y})^T}_{\boldsymbol{V}^T}.$$

3 Principal Component Analysis (PCA)

3.1 Data Matrix and Basic Statistics

Suppose we have a dataset consisting of m *samples* (observations, rows) and n *features* (variables/attributes, columns), stored in a raw data matrix $\widetilde{\boldsymbol{X}} \in \mathbb{R}^{m \times n}$:

$$\widetilde{\boldsymbol{X}} = \begin{pmatrix} — & \tilde{\boldsymbol{x}}_1^T & — \\ — & \tilde{\boldsymbol{x}}_2^T & — \\ & \vdots & \\ — & \tilde{\boldsymbol{x}}_m^T & — \end{pmatrix} \in \mathbb{R}^{m \times n}.$$

To review basic sample statistics, suppose we have two features (columns of $\widetilde{\boldsymbol{X}}$), denoted $\boldsymbol{a} = (a_1, a_2, \ldots, a_m)^T$ (e.g., height) and $\boldsymbol{b} = (b_1, b_2, \ldots, b_m)^T$ (e.g., weight), so each sample is a row vector $\tilde{\boldsymbol{x}}_i^T = (a_i, b_i)$:

1. **Sample Mean (Estimate of Expectation):**

$$\mu_A = \frac{1}{m} \sum_{i=1}^{m} a_i.$$

2. **Sample Variance:** Measures the spread of feature A around its mean:

$$\mathrm{Var}(A) = \frac{1}{m-1} \sum_{i=1}^{m} (a_i - \mu_A)^2 = \frac{1}{m-1} \left((a_1 - \mu_A)^2 + \cdots + (a_m - \mu_A)^2 \right),$$

where dividing by $m-1$ instead of m is *Bessel's correction*, which provides an unbiased estimator of the population variance when the mean μ_A is estimated from the same sample.

3. **Sample Covariance:** Measures the joint linear variability of two features A and B:

$$\mathrm{Cov}(A, B) = \frac{1}{m-1} \sum_{i=1}^{m} (a_i - \mu_A)(b_i - \mu_B).$$

Note that $\mathrm{Cov}(A, A) = \mathrm{Var}(A)$ and $\mathrm{Cov}(A, B) = \mathrm{Cov}(B, A)$.

3.2 Centering the Data and the Covariance Matrix

Let $\boldsymbol{\mu} \in \mathbb{R}^{1\times n}$ be the row vector of column means:

$$\boldsymbol{\mu} = \left(\frac{1}{m}\sum_{i=1}^{m} \tilde{x}_{i,1},\ \frac{1}{m}\sum_{i=1}^{m} \tilde{x}_{i,2},\ \ldots,\ \frac{1}{m}\sum_{i=1}^{m} \tilde{x}_{i,n} \right).$$

We **center** the dataset by subtracting the mean row $\boldsymbol{\mu}$ from every row of $\widetilde{\boldsymbol{X}}$:

$$\boldsymbol{X} = \widetilde{\boldsymbol{X}} - \begin{pmatrix} \boldsymbol{\mu} \\ \boldsymbol{\mu} \\ \vdots \\ \boldsymbol{\mu} \end{pmatrix} \in \mathbb{R}^{m\times n},$$

so that each column of $\boldsymbol{X}$ has zero mean.

The **sample covariance matrix** $\boldsymbol{C} \in \mathbb{R}^{n\times n}$ of the centered data matrix $\boldsymbol{X}$ is then given concisely by

$$\boldsymbol{C} = \frac{1}{m-1}\boldsymbol{X}^T\boldsymbol{X} = \begin{pmatrix} \mathrm{Cov}(A,A) & \mathrm{Cov}(A,B) & \cdots \\ \mathrm{Cov}(B,A) & \mathrm{Cov}(B,B) & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}. \tag{8}$$

Here the diagonal entries $C_{jj} = \mathrm{Var}(\text{feature } j)$ are the variances of the individual features, and the off-diagonal entries C_{jk} are the covariances between pairs of features.

3.3 PCA via Eigen-Decomposition and SVD

Since $\boldsymbol{C} = \frac{1}{m-1}\boldsymbol{X}^T\boldsymbol{X}$ is symmetric positive semi-definite, it admits an orthogonal eigenvalue decomposition:

$$\boldsymbol{C} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^T, \qquad \boldsymbol{\Lambda} = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n), \quad \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0. \tag{9}$$

Now consider the Singular Value Decomposition of the centered data matrix $\boldsymbol{X}$:

$$\boldsymbol{X} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T.$$

Substituting $\boldsymbol{X} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$ into the covariance matrix (8) gives

$$\boldsymbol{C} = \frac{1}{m-1}\boldsymbol{X}^T\boldsymbol{X} = \frac{1}{m-1}\boldsymbol{V}\boldsymbol{\Sigma}^T\underbrace{\boldsymbol{U}^T\boldsymbol{U}}_{\boldsymbol{I}}\boldsymbol{\Sigma}\boldsymbol{V}^T = \boldsymbol{V}\left(\frac{\boldsymbol{\Sigma}^2}{m-1}\right)\boldsymbol{V}^T. \tag{10}$$

Comparing $\boldsymbol{C} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^T$ with $\boldsymbol{C} = \boldsymbol{V}\frac{\boldsymbol{\Sigma}^2}{m-1}\boldsymbol{V}^T$ reveals the direct connection between PCA and SVD:

1. **Principal Axes (Directions):** The right singular vectors of $\boldsymbol{X}$ (columns of $\boldsymbol{V} = (\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n)$) are the eigenvectors of the covariance matrix $\boldsymbol{C}$. They define the new orthonormal basis of uncorrelated directions maximizing variance.

2. **Eigenvalues (Variances along Principal Axes):** The eigenvalues of $\boldsymbol{C}$ are related to the singular values of $\boldsymbol{X}$ by

$$\lambda_i = \frac{\sigma_i^2}{m-1}.$$

3. **Principal Components (Projected Data / Scores):** Multiplying $\boldsymbol{X} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$ on the right by $\boldsymbol{V}$ gives

$$\boxed{\boldsymbol{X}\boldsymbol{V} = \boldsymbol{U}\boldsymbol{\Sigma} = (\sigma_1\boldsymbol{u}_1,\ \sigma_2\boldsymbol{u}_2,\ \ldots,\ \sigma_n\boldsymbol{u}_n).}$$

The matrix $\boldsymbol{X}\boldsymbol{V} = \boldsymbol{U}\boldsymbol{\Sigma} \in \mathbb{R}^{m\times n}$ represents the **change of basis** of the rows of $\boldsymbol{X}$ into the orthonormal basis formed by the columns of $\boldsymbol{V}$:

- Each **row** j of $\boldsymbol{X}\boldsymbol{V} = \boldsymbol{U}\boldsymbol{\Sigma}$ is the j-th data sample $\boldsymbol{x}_j^T\boldsymbol{V}$ expressed in the new coordinates (the principal axes).
- Each **column** i of $\boldsymbol{X}\boldsymbol{V} = \boldsymbol{U}\boldsymbol{\Sigma}$ is $\boldsymbol{X}\boldsymbol{v}_i = \sigma_i\boldsymbol{u}_i \in \mathbb{R}^m$ (the i-th column of $\boldsymbol{U}$ *scaled by* σ_i), which gives the projections (scores) of all m samples onto the i-th principal axis $\boldsymbol{v}_i$; these columns $\sigma_i\boldsymbol{u}_i$ are called the *principal components*.

To reduce dimensionality from n features to $k < n$ features, we keep only the first k columns:

$$\boldsymbol{X}_k = \boldsymbol{X}\boldsymbol{V}_k = \boldsymbol{U}_k\boldsymbol{\Sigma}_k = (\sigma_1\boldsymbol{u}_1, \ldots, \sigma_k\boldsymbol{u}_k) \in \mathbb{R}^{m\times k}.$$

3.4 Total Variance and Explained Variance Ratio

Since the trace of a matrix is invariant under orthogonal similarity transformations ($\mathrm{tr}(\boldsymbol{C}) = \mathrm{tr}(\boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^T) = \mathrm{tr}(\boldsymbol{\Lambda})$), the **total variance** T of the dataset equals both the sum of the original feature variances and the sum of the eigenvalues of $\boldsymbol{C}$:

$$T = \mathrm{tr}(\boldsymbol{C}) = \sum_{i=1}^{n} \mathrm{Var}(\text{feature } i) = \sum_{i=1}^{n} \lambda_i = \frac{1}{m-1}\sum_{i=1}^{n} \sigma_i^2. \tag{11}$$

The fraction of the total variance captured by the i-th principal component $\boldsymbol{v}_i$ is the **explained variance ratio**:

$$\frac{\lambda_i}{T} = \frac{\lambda_i}{\sum_{j=1}^{n}\lambda_j} = \frac{\sigma_i^2}{\sum_{j=1}^{n}\sigma_j^2}.$$

Example 3.1 (2D Covariance Matrices and Principal Axes)**.** Consider two 2D datasets with total variance $T = 100$:

1. **Nearly uncorrelated features aligned with coordinate axes:**

$$\boldsymbol{C}_1 = \begin{pmatrix} 95 & 1 \\ 1 & 5 \end{pmatrix}.$$

Here the data spread is almost entirely along the horizontal axis:

$$\lambda_1 \approx 95, \quad \lambda_2 \approx 5, \qquad \boldsymbol{v}_1 \approx \begin{pmatrix} 0.999 \\ 0.011 \end{pmatrix}, \quad \boldsymbol{v}_2 \approx \begin{pmatrix} -0.011 \\ 0.999 \end{pmatrix}.$$

The first principal component $\boldsymbol{v}_1$ captures $\lambda_1/T \approx 95\%$ of the total variance.

2. **Strongly correlated features tilted at** $45°$**:**

$$\boldsymbol{C}_2 = \begin{pmatrix} 50 & 40 \\ 40 & 50 \end{pmatrix}.$$

Each individual coordinate has variance 50, but they covary strongly (Cov = 40). Computing the eigenvalues via $\det(\boldsymbol{C}_2 - \lambda\boldsymbol{I}) = 0$:

$$\det(\boldsymbol{C}_2 - \lambda\boldsymbol{I}) = \begin{vmatrix} 50-\lambda & 40 \\ 40 & 50-\lambda \end{vmatrix} = (50-\lambda)^2 - 40^2 = 0 \quad \Longrightarrow \quad 50 - \lambda = \pm 40,$$

which gives:

$$\lambda_1 = 90, \quad \boldsymbol{v}_1 = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}; \qquad \lambda_2 = 10, \quad \boldsymbol{v}_2 = \begin{pmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}.$$

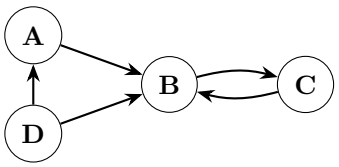
The total variance is $T = 50 + 50 = \lambda_1 + \lambda_2 = 90 + 10 = 100$. Projecting onto the 45° diagonal $\boldsymbol{v}_1$ captures $\frac{\lambda_1}{T} = \frac{90}{100} = 90\%$ of the total variance in a single coordinate!

4 PageRank Algorithm

4.1 The Web Graph and Link Democracy

PageRank models the World Wide Web as a directed graph $G = (V, E)$ with $N = |V|$ pages (nodes), where a directed edge $y \to x$ represents a hyperlink from page y to page x.

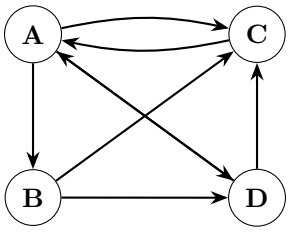
- **Link Democracy:** A hyperlink pointing to a page is treated as a "vote" of confidence or trust. We can represent the directed web graph by an adjacency matrix $\boldsymbol{A} \in \{0,1\}^{N \times N}$ where $A_{ij} = 1$ if page j links to page i ($j \to i$) and 0 otherwise. For instance, on the 4-node graph with edges $A \to B$, $B \leftrightarrow C$, $D \to A$, and $D \to B$:



$$\boldsymbol{A} = \begin{array}{c} \\ A \\ B \\ C \\ D \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & D \end{array} \\ \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \end{array}$$

- **Recursive Importance:** Not all votes are equal—a link from an important page carries more weight than a link from an obscure page. Furthermore, a page y that links to $\text{out}(y)$ pages splits its vote equally among all its outgoing links.

Example 4.1 (4-Node Web Graph and Transition Matrix)**.** Consider a 4-page web graph with nodes A, B, C, D:



Here:

- Node A has $\text{out}(A) = 3$ outgoing links ($A \to B, C, D$).
- Node B has $\text{out}(B) = 2$ outgoing links ($B \to C, D$).
- Node C has $\text{out}(C) = 1$ outgoing link ($C \to A$).
- Node D has $\text{out}(D) = 2$ outgoing links ($D \to A, C$).

The ideal recursive PageRank equations are:

$$\begin{aligned} PR(A) &= PR(C) + \frac{PR(D)}{2}, \\ PR(B) &= \frac{PR(A)}{3}, \\ PR(C) &= \frac{PR(A)}{3} + \frac{PR(B)}{2} + \frac{PR(D)}{2}, \\ PR(D) &= \frac{PR(A)}{3} + \frac{PR(B)}{2}. \end{aligned}$$

In matrix form, $\boldsymbol{r} = \boldsymbol{M}\boldsymbol{r}$ (an eigenvalue problem $\boldsymbol{M}\boldsymbol{r} = 1 \cdot \boldsymbol{r}$ for eigenvalue $\lambda = 1$), where the column-stochastic transition (Markov) matrix $\boldsymbol{M}$ is:

$$\boldsymbol{M} = \begin{pmatrix} 0 & 0 & 1 & 1/2 \\ 1/3 & 0 & 0 & 0 \\ 1/3 & 1/2 & 0 & 1/2 \\ 1/3 & 1/2 & 0 & 0 \end{pmatrix}.$$

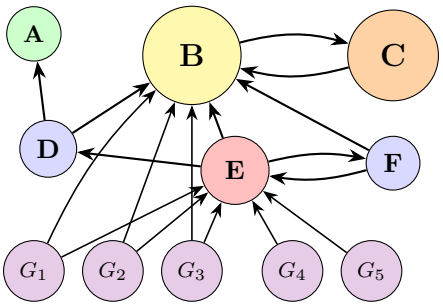
4.2 Random Surfer Model and Teleportation (Damping Factor)

Consider a **random surfer** who starts on a random web page and clicks outgoing links uniformly at random infinitely many times. If $\boldsymbol{p}(t)$ is the probability distribution vector of the surfer's location at step t, then

$$\boldsymbol{p}(t+1) = \boldsymbol{M}\boldsymbol{p}(t).$$

As $t \to \infty$, we seek a *steady state* (stationary distribution) $\boldsymbol{p}(t+1) = \boldsymbol{p}(t) = \boldsymbol{r}$, where $PR(x) = r_x$ is the probability of finding the surfer at page x at a random time.

Example 4.2 (Random Surfer Walk and Pathologies on the Wikimedia Example Graph[3])**.** Consider the classic 11-node web graph consisting of pages A, B, C, D, E, F and five pages $G_1, \ldots, G_5$ that have no incoming links:



Follow a random surfer starting at one of the bottom pages G:

$$G \to E \to F \to E \to D \to B \to C \quad \Longrightarrow \quad B \leftrightarrow C \text{ forever!}$$

This graph illustrates two fundamental pathologies of pure link-following ($\boldsymbol{p}(t+1) = \boldsymbol{M}\boldsymbol{p}(t)$):

1. **Spider traps (cycles):** Once the surfer enters B or C, the closed cycle $B \leftrightarrow C$ traps them forever.

[3]Graph from https://commons.wikimedia.org/wiki/File:PageRanks-Example.svg (see also `108/pagerank.py`).

2. **Sink (dangling) nodes:** Page A has no outgoing links ($\text{out}(A) = 0$), corresponding to a zero column in the raw adjacency matrix.

To resolve spider traps and ensure a unique stationary distribution, Brin and Page introduced a **damping factor** $\lambda \in (0, 1)$ (typically $\lambda = 0.85$):

- With probability λ, the surfer follows an outgoing link uniformly at random.
- With probability $1 - \lambda$, the surfer gets bored and **teleports** (jumps) to a completely random page chosen uniformly among all N pages.

This yields the damped PageRank formula for each page x:

$$PR(x) = \frac{1-\lambda}{N} + \lambda \sum_{y \to x} \frac{PR(y)}{\text{out}(y)}. \tag{12}$$

4.3 Power Iteration Algorithm and Handling Sink Nodes

Equation (12) is solved iteratively via the **Power Method**:

1. **Initialize:** Set $PR_0(x) = \frac{1}{N}$ for all $x = 1, \ldots, N$ (e.g., $PR_0(x) = \frac{1}{11}$ on the 11-node Wikimedia graph).
2. **Iterate until convergence:** For $k = 0, 1, 2, \ldots$:

$$\forall x: \quad PR_{k+1}(x) = \frac{1-\lambda}{N} + \lambda \sum_{y \to x} \frac{PR_k(y)}{\text{out}(y)}.$$

Example 4.3 (First Iterations on the 11-Node Graph: Rapid Convergence)**.** Consider the 11-node Wikimedia graph from Example 4.2 with $N = 11$, $\lambda = 0.85$ ($1 - \lambda = 0.15$), and uniform initial rank $PR_0(x) = \frac{1}{11} \approx 0.0909$ for all pages:

- **Iteration 1 for node B:** Page B receives links from C ($\text{out}(C) = 1$), D ($\text{out}(D) = 2$), E ($\text{out}(E) = 3$), F ($\text{out}(F) = 2$), and G_1, G_2, G_3 ($\text{out}(G_i) = 2$ each):

$$\begin{aligned} PR_1(B) &= 0.15 \cdot \frac{1}{11} + 0.85 \left(PR_0(C) + \frac{PR_0(D)}{2} + \frac{PR_0(E)}{3} + \frac{PR_0(F)}{2} + 3 \cdot \frac{PR_0(G_1)}{2} \right) \\ &= \frac{0.15}{11} + 0.85 \cdot \frac{1}{11} \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{2} + \frac{3}{2} \right) \approx 0.31. \end{aligned}$$

  In a single step, B's rank jumps from $\frac{1}{11} \approx 0.09$ to 0.31 (already close to its exact stationary value 0.384).

- **Iteration 1 for node C:** Page C only receives a link from B ($\text{out}(B) = 1$), which at $k = 0$ still has $PR_0(B) = \frac{1}{11}$:

$$PR_1(C) = \frac{0.15}{11} + 0.85 \left(\frac{1}{11} \right) = \frac{1}{11} \approx 0.0909.$$

- **Iteration 2 for node C:** In the second step, B's high rank $PR_1(B) \approx 0.31$ propagates directly to C:

$$PR_2(C) = \frac{0.15}{11} + 0.85 \cdot PR_1(B) \approx \frac{0.15}{11} + 0.85 \times 0.31 \approx 0.28.$$

  Thus, after just **two iterations**, $PR_2(C) \approx 0.28$ has already surged near its final converged value (0.343), illustrating the extremely fast initial convergence of PageRank power iteration (in `108/pagerank.py`, the ℓ_1-norm change drops below 0.1 in just 10 iterations).

Problem with Sink Nodes: Suppose we have a 2-node graph $A \to B$ where B is a sink node ($\text{out}(B) = 0$) and $\lambda = 0.8$. Starting with $PR(A) = PR(B) = 0.5$:

- For node A (no incoming links): $PR_1(A) = \frac{1-\lambda}{2} + \lambda \cdot 0 = \frac{0.2}{2} = 0.1$.
- For node B (link from A with $\text{out}(A) = 1$):
  - Using $PR_0(A) = 0.5$: $PR_1(B) = 0.1 + 0.8 \times 0.5 = 0.5$, and in the next step $PR_2(B) = 0.1 + 0.8 \times 0.1 = 0.18$.
  - Using the updated $PR_1(A) = 0.1$ immediately: $PR(B) = 0.1 + 0.8 \times 0.1 = 0.18$.

In either case, at steady state $PR(A) = 0.1$ and $PR(B) = 0.18$, so $PR(A) + PR(B) = 0.28 \neq 1$! Probability mass leaks out of the system through the sink node B.

Solution — Random Hop from Sink Nodes: When the surfer reaches a sink node z ($z \to \emptyset$), they teleport uniformly at random to any page in the web with probability $1/N$. Conceptually, this connects every sink node to all N nodes, which would make the matrix $\boldsymbol{M}$ dense. However, we can compute this **without forming a dense matrix** by accumulating the total PageRank of all sink nodes into a single scalar S:

$$PR(x) = \frac{1-\lambda}{N} + \lambda \sum_{y \to x} \frac{PR(y)}{\text{out}(y)} + \lambda \sum_{z \to \emptyset} \frac{PR(z)}{N} = \frac{1 - \lambda + \lambda S}{N} + \lambda \sum_{y \to x} \frac{PR(y)}{\text{out}(y)}, \quad S = \sum_{z \to \emptyset} PR(z). \tag{13}$$

4.4 Matrix Formulation and Personalized PageRank

Assuming sink columns in $\boldsymbol{M}$ are replaced by $\frac{1}{N}\boldsymbol{e}$ (so $\boldsymbol{M}$ is column-stochastic), the PageRank vector $\boldsymbol{r} = (PR(1), \ldots, PR(N))^T$ satisfies:

$$\boldsymbol{r} = \lambda \boldsymbol{M}\boldsymbol{r} + \frac{1-\lambda}{N}\boldsymbol{e}, \qquad \text{where } \boldsymbol{e} = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} \in \mathbb{R}^N. \tag{14}$$

Example 4.4 (Computed PageRank on the 11-Node Wikimedia Graph)**.** Applying the damped PageRank iteration ($\lambda = 0.85$, with sink node A teleporting uniformly to all $N = 11$ nodes) to the 11-node Wikimedia graph from Example 4.2 yields the stationary probabilities (expressed as percentages):

Node	B	C	E	D	F	A	$G_1, \ldots, G_5$ (each)
In-degree	7	1	6	1	1	1	0
PageRank	**38.4**%	**34.3**%	8.1%	3.9%	3.9%	3.3%	1.6%

Notice how this illustrates *recursive importance*:

- Page E has 6 incoming links (from F and $G_1, \ldots, G_5$), yet its PageRank is only 8.1% because $G_1, \ldots, G_5$ have no incoming links of their own (their PageRank is purely the minimum teleportation floor $\frac{1-0.85}{11} + 0.85\frac{PR(A)}{11} \approx 1.62\%$).
- Page C has only **one** incoming link, but that link comes from page B (38.4%), which has no other outgoing links ($\text{out}(B) = 1$). Consequently, C inherits $\lambda \cdot PR(B) = 0.85 \times 38.4\%$ of B's rank, giving $PR(C) = 34.3\%$!

Personalized PageRank (Recommender Systems & Topic-Sensitive Search): Instead of teleporting uniformly to all pages in the web via $\frac{1}{N}\boldsymbol{e}$, we can restrict random jumps to a specific subset of interest (e.g., pages in an online encyclopedia, a user's bookmarked pages, or items a user has interacted with in a recommender system). Replacing $\frac{1}{N}\boldsymbol{e}$ with a non-negative personalization vector $\boldsymbol{c}$ (with $\sum_i c_i = 1$, or a binary indicator vector normalized by the number of selected nodes) gives:

$$\boldsymbol{r} = \lambda \boldsymbol{M}\boldsymbol{r} + (1-\lambda)\tilde{\boldsymbol{c}} \qquad \left(\text{or } \boldsymbol{r} = \lambda \boldsymbol{M}\boldsymbol{r} + \frac{1-\lambda}{|\mathcal{S}|}\boldsymbol{c} \text{ for } c_i \in \{0,1\}\right). \tag{15}$$

5 Sparse Direct Solvers and Matrix Reorderings

5.1 Block LU Factorization, Fill-In, and Elimination Graphs

Consider the first step of Gaussian elimination (LU factorization) on a sparse matrix $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ partitioned into its top-left pivot $\alpha = a_{11} \neq 0$, first row $\boldsymbol{r} \in \mathbb{R}^{1\times(n-1)}$, first column $\boldsymbol{c} \in \mathbb{R}^{(n-1)\times 1}$, and trailing submatrix $\boldsymbol{B} \in \mathbb{R}^{(n-1)\times(n-1)}$:

$$\boldsymbol{A} = \begin{pmatrix} \alpha & \boldsymbol{r} \\ \boldsymbol{c} & \boldsymbol{B} \end{pmatrix} = \begin{pmatrix} 1 & \boldsymbol{0} \\ \boldsymbol{c}/\alpha & \boldsymbol{I} \end{pmatrix} \begin{pmatrix} \alpha & \boldsymbol{r} \\ \boldsymbol{0} & \boldsymbol{C} \end{pmatrix} = \boldsymbol{L}\boldsymbol{U}, \tag{16}$$

where the remaining $(n-1)\times(n-1)$ block to be factored is the **Schur complement**:

$$\boldsymbol{C} = \boldsymbol{B} - \frac{1}{\alpha}\boldsymbol{c}\boldsymbol{r}. \tag{17}$$

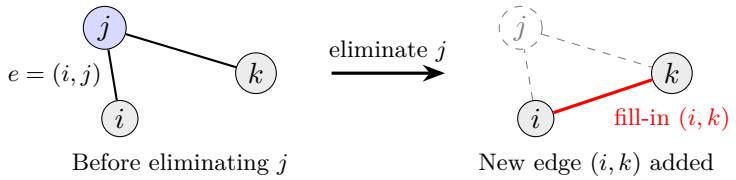
If $\alpha = 0$ (or is very small), row permutations $\boldsymbol{P}_r\boldsymbol{A} = \boldsymbol{L}\boldsymbol{U}$ (pivoting) are required for numerical stability.

Fill-in: Notice that the rank-1 outer product $\boldsymbol{c}\boldsymbol{r}/\alpha$ modifies entry (i,k) whenever $c_i = a_{i1} \neq 0$ and $r_k = a_{1k} \neq 0$. More generally, when eliminating pivot row/column j:

$$\begin{pmatrix} a_{jj} & a_{jk} \\ a_{ij} & 0 \end{pmatrix} \xrightarrow{\text{eliminate } j} \begin{pmatrix} a_{jj} & a_{jk} \\ 0 & F = \tilde{a}_{ik} = -\frac{a_{ij}a_{jk}}{a_{jj}} \neq 0 \end{pmatrix}.$$

A previously zero entry $a_{ik} = 0$ becomes a non-zero entry $F \neq 0$; this new non-zero is called **fill-in**.

Graph Interpretation: Associate with the sparse symmetric matrix $\boldsymbol{A}$ an undirected *adjacency graph* $G = (V, E)$, where vertex $i \in \{1, \ldots, n\}$ corresponds to row/column i, and an edge $(i,j) \in E$ exists if and only if $a_{ij} \neq 0$.



Elimination Rule: When vertex j is eliminated, vertex j and its incident edges are removed, and **new edges (fill-in) are added between all pairs of neighbors of j that were not already connected** (i.e., the neighbors of j form a clique).

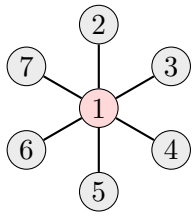
Example 5.1 (Fill-In in the 2D Poisson Matrix $\boldsymbol{K}_{2\mathrm{D}}$)**.** For the 2D Poisson problem on an $N \times N$ grid ordered row-by-row $(1, 2, 3, 4 \mid 5, 6, 7, 8 \mid 9, 10, 11, 12 \dots)$, the 5-point stencil produces a block tridiagonal matrix:

$$\boldsymbol{K}_{2\mathrm{D}} = \begin{pmatrix} \boldsymbol{K}+2\boldsymbol{I} & -\boldsymbol{I} & & \\ -\boldsymbol{I} & \boldsymbol{K}+2\boldsymbol{I} & -\boldsymbol{I} & \\ & \ddots & \ddots & \ddots \\ & & -\boldsymbol{I} & \boldsymbol{K}+2\boldsymbol{I} \end{pmatrix}, \qquad \boldsymbol{K} = \begin{pmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & \ddots & \ddots & \ddots \end{pmatrix}.$$

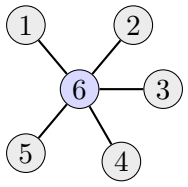
Between the main tridiagonal band and the outer $-\boldsymbol{I}$ bands (which are at distance N from the diagonal), all entries are initially zero. However, during LU factorization, eliminating row 1 creates fill-in F between its right neighbor (2) and its top grid neighbor $(N+1)$, which then cascades across the entire band between distance 1 and distance N:

$$\begin{pmatrix} 4 & -1 & & & -1 & & & \\ -1 & 4 & -1 & & F & -1 & & \\ & -1 & 4 & -1 & F & F & -1 & \\ & & -1 & 4 & \cdots & \cdots & \cdots & -1 \\ -1 & F & F & \cdots & 4 & -1 & & \end{pmatrix}.$$

Example 5.2 (Star Graph: Arrowhead vs. Hub-Last Ordering)**.** Consider a star graph where a central hub node is connected to leaf nodes:



(a) Hub numbered first (1)



(b) Hub numbered last (6)

- In ordering (a), where the hub is numbered 1, the first row and column are completely dense:

$$\boldsymbol{A}_1 = \begin{pmatrix} \times & \times & \times & \times & \times & \times \\ \times & \times & 0 & 0 & 0 & 0 \\ \times & 0 & \times & 0 & 0 & 0 \\ \times & 0 & 0 & \times & 0 & 0 \\ \times & 0 & 0 & 0 & \times & 0 \\ \times & 0 & 0 & 0 & 0 & \times \end{pmatrix}.$$

  Eliminating node 1 connects all of its neighbors $(2, \dots, 6)$, turning the remaining submatrix into a completely dense clique (**100**% fill-in)!

- In ordering (b), obtained by symmetrically permuting rows and columns via $\boldsymbol{P}\boldsymbol{A}\boldsymbol{P}^T$ so that the leaves are numbered $1, \dots, 5$ and the hub is numbered last (6), the matrix has an upward-pointing arrowhead structure:

$$\boldsymbol{P}\boldsymbol{A}\boldsymbol{P}^T = \begin{pmatrix} \times & 0 & 0 & 0 & 0 & \times \\ 0 & \times & 0 & 0 & 0 & \times \\ 0 & 0 & \times & 0 & 0 & \times \\ 0 & 0 & 0 & \times & 0 & \times \\ 0 & 0 & 0 & 0 & \times & \times \\ \times & \times & \times & \times & \times & \times \end{pmatrix} \xrightarrow{\mathrm{LU}} \textbf{zero fill-in!}$$

  Because each leaf $1, \dots, 5$ has only a single neighbor (node 6), eliminating them introduces no new edges at all.

5.2 Reordering Algorithms

To minimize fill-in and factorization cost in sparse direct solvers, we seek a permutation matrix $\boldsymbol{P}$ and solve $(\boldsymbol{PAP}^T)(\boldsymbol{Px}) = \boldsymbol{Pb}$.

1. **Red-Black (Checkerboard) Ordering:** Color the nodes of a 2D grid like a checkerboard into two independent sets: Red (R) and Black (B), where no two nodes of the same color share an edge (every Red node only connects to Black neighbors, and vice versa). Numbering all Red nodes first $(1, 2, 3, 4)$ and all Black nodes second $(5, 6, 7, 8)$ permutes the 2D Poisson matrix into a 2×2 block form:

$$\boldsymbol{PK}_{\text{2D}}\boldsymbol{P}^T = \begin{pmatrix} 4\boldsymbol{I}_R & \boldsymbol{C} \\ \boldsymbol{C}^T & 4\boldsymbol{I}_B \end{pmatrix}.$$

   Since the top-left block $4\boldsymbol{I}_R$ is strictly diagonal, all Red unknowns can be eliminated in parallel in a single step!

2. **Minimum Degree (MD / AMD / QMD):** A greedy local heuristic inspired by the star-graph example:

   - At each step, select and eliminate the vertex with the **smallest current degree** (fewest neighbors $\implies$ smallest clique of fill-in edges added).
   - Update the elimination graph (remove the vertex, connect its neighbors into a clique) and update vertex degrees.
   - Practical implementations use *Quotient Minimum Degree (QMD)* or *Approximate Minimum Degree (AMD)* to update degrees without explicitly forming all fill-in edges.

3. **Nested Dissection (ND):** A global divide-and-conquer strategy:

   - Find a small vertex **separator** S that splits the graph into two disconnected subgraphs P and Q of roughly equal size (no direct edges between P and Q).
   - Number the nodes in P first, then the nodes in Q, and place the separator nodes S **last**:

$$\boldsymbol{PAP}^T = \begin{pmatrix} \boldsymbol{A}_{PP} & \boldsymbol{O} & \boldsymbol{A}_{PS} \\ \boldsymbol{O} & \boldsymbol{A}_{QQ} & \boldsymbol{A}_{QS} \\ \boldsymbol{A}_{SP} & \boldsymbol{A}_{SQ} & \boldsymbol{A}_{SS} \end{pmatrix}.$$

   - Because of the $\boldsymbol{O}$ blocks, eliminating P and Q produces zero fill-in between P and Q and can be done completely in parallel. Applying this recursively to P and Q yields optimal asymptotic complexity for 2D/3D mesh problems.

4. **(Reverse) Cuthill–McKee (CM / RCM):** Designed to reduce the **bandwidth** of $\boldsymbol{A}$ (clustering all non-zeros close to the main diagonal):

   (a) Pick a starting vertex with minimum degree (or a peripheral node).

   (b) Perform **Breadth-First Search (BFS)** level by level.

   (c) Within each level, sort the newly visited neighbors by **increasing degree**.

   (d) **RCM:** Reverse the resulting ordering at the end (which further reduces profile/envelope fill-in when high-degree nodes are pushed toward the end).

6 Quadratic Function Minimization and GMRES

6.1 Quadratic Cost Function and Level Sets

When $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ is symmetric positive definite ($\boldsymbol{A} = \boldsymbol{A}^T \succ 0$), solving $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ is equivalent to minimizing the strictly convex quadratic function

$$f(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{x}^T\boldsymbol{A}\boldsymbol{x} - \boldsymbol{x}^T\boldsymbol{b}. \tag{18}$$

Let $\boldsymbol{x}^* = \boldsymbol{A}^{-1}\boldsymbol{b}$ be the exact solution ($\boldsymbol{A}\boldsymbol{x}^* - \boldsymbol{b} = \boldsymbol{0}$), and let $\boldsymbol{y} = \boldsymbol{x} - \boldsymbol{x}^*$ ($\iff$ $\boldsymbol{x} = \boldsymbol{y} + \boldsymbol{x}^*$) be the error vector. Expanding $f(\boldsymbol{y} + \boldsymbol{x}^*)$ around $\boldsymbol{x}^*$ gives:

$$\begin{aligned} f(\boldsymbol{y} + \boldsymbol{x}^*) &= \frac{1}{2}(\boldsymbol{y} + \boldsymbol{x}^*)^T\boldsymbol{A}(\boldsymbol{y} + \boldsymbol{x}^*) - (\boldsymbol{y} + \boldsymbol{x}^*)^T\boldsymbol{b} \\ &= \frac{1}{2}\boldsymbol{y}^T\boldsymbol{A}\boldsymbol{y} + \boldsymbol{y}^T\underbrace{(\boldsymbol{A}\boldsymbol{x}^* - \boldsymbol{b})}_{\boldsymbol{0}} + \underbrace{\left[\frac{1}{2}(\boldsymbol{x}^*)^T\boldsymbol{A}\boldsymbol{x}^* - (\boldsymbol{x}^*)^T\boldsymbol{b}\right]}_{f(\boldsymbol{x}^*)} \\ &= \frac{1}{2}\boldsymbol{y}^T\boldsymbol{A}\boldsymbol{y} + f(\boldsymbol{x}^*). \end{aligned}$$

Now consider a **level set** (contour surface) $f(\boldsymbol{x}) = c$ for some constant $c > f(\boldsymbol{x}^*)$:

$$f(\boldsymbol{x}) = c \quad\iff\quad \frac{1}{2}\boldsymbol{y}^T\boldsymbol{A}\boldsymbol{y} = c - f(\boldsymbol{x}^*) =: k \quad\iff\quad \boldsymbol{y}^T\boldsymbol{A}\boldsymbol{y} = 2k.$$

Using the orthogonal spectral decomposition $\boldsymbol{A} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^T$ and changing coordinates to $\boldsymbol{z} = \boldsymbol{V}^T\boldsymbol{y}$ (aligned with the eigenvectors of $\boldsymbol{A}$), the level set equation becomes:

$$\boldsymbol{y}^T\boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^T\boldsymbol{y} = \boldsymbol{z}^T\boldsymbol{\Lambda}\boldsymbol{z} = 2k.$$

In 2D ($\boldsymbol{z} = (z_1, z_2)^T$), this is the equation of an **ellipse**:

$$\lambda_1 z_1^2 + \lambda_2 z_2^2 = 2k \quad\iff\quad \underbrace{\frac{z_1^2}{(2k/\lambda_1)}}_{\alpha^2} + \underbrace{\frac{z_2^2}{(2k/\lambda_2)}}_{\beta^2} = 1, \tag{19}$$

with semi-axes of lengths $\alpha = \sqrt{\frac{2k}{\lambda_1}}$ and $\beta = \sqrt{\frac{2k}{\lambda_2}}$ directed along the eigenvectors of $\boldsymbol{A}$.

- Larger eigenvalues λ_i correspond to *shorter* semi-axes $\sqrt{2k/\lambda_i}$ (steeper curvature).
- The ratio of the longest to the shortest semi-axis of the contour ellipsoid is $\sqrt{\lambda_{\max}/\lambda_{\min}} = \sqrt{\kappa_2(\boldsymbol{A})}$. When $\kappa_2(\boldsymbol{A}) \gg 1$, the level sets are highly elongated valleys, making gradient descent methods zig-zag and converge slowly.

6.2 Krylov Subspace Minimization and the GMRES Idea

Now let $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ be a general (possibly non-symmetric) invertible matrix, and let $\bar{\boldsymbol{x}} = \boldsymbol{A}^{-1}\boldsymbol{b}$ denote the exact solution of $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$. At step k, let $\boldsymbol{e}_k = \boldsymbol{x}_k - \bar{\boldsymbol{x}}$ be the error and $\boldsymbol{r}_k = \boldsymbol{b} - \boldsymbol{A}\boldsymbol{x}_k = -\boldsymbol{A}\boldsymbol{e}_k$ be the residual. Notice that minimizing the Euclidean norm of the residual $\|\boldsymbol{r}_k\|_2$ is equivalent to minimizing the $\boldsymbol{A}^T\boldsymbol{A}$-norm of the error:

$$\|\boldsymbol{e}_k\|^2_{\boldsymbol{A}^T\boldsymbol{A}} = \|\boldsymbol{x}_k - \bar{\boldsymbol{x}}\|^2_{\boldsymbol{A}^T\boldsymbol{A}} = \boldsymbol{e}_k^T\boldsymbol{A}^T\boldsymbol{A}\boldsymbol{e}_k = (\boldsymbol{A}\boldsymbol{e}_k)^T(\boldsymbol{A}\boldsymbol{e}_k) = (\boldsymbol{A}\boldsymbol{x}_k - \boldsymbol{b})^T(\boldsymbol{A}\boldsymbol{x}_k - \boldsymbol{b}) = \|\boldsymbol{r}_k\|_2^2.$$

The **Generalized Minimal Residual method (GMRES)** finds the iterate $\boldsymbol{x}_k$ in the affine Krylov subspace $\boldsymbol{x}_0 + \mathcal{K}_k(\boldsymbol{A}, \boldsymbol{r}_0)$ that minimizes the residual 2-norm:

$$\boldsymbol{x}_k = \underset{\boldsymbol{x} \in \boldsymbol{x}_0 + \mathcal{K}_k(\boldsymbol{A}, \boldsymbol{r}_0)}{\arg\min} \|\boldsymbol{b} - \boldsymbol{A}\boldsymbol{x}\|_2, \qquad \mathcal{K}_k(\boldsymbol{A}, \boldsymbol{r}_0) = \operatorname{span}\{\boldsymbol{r}_0, \boldsymbol{A}\boldsymbol{r}_0, \boldsymbol{A}^2\boldsymbol{r}_0, \ldots, \boldsymbol{A}^{k-1}\boldsymbol{r}_0\}. \tag{20}$$

Let $\boldsymbol{K}_k = (\boldsymbol{r}_0, \boldsymbol{A}\boldsymbol{r}_0, \boldsymbol{A}^2\boldsymbol{r}_0, \ldots, \boldsymbol{A}^{k-1}\boldsymbol{r}_0) \in \mathbb{R}^{n \times k}$ be the Krylov matrix. For $\boldsymbol{x}_0 = \boldsymbol{0}$, any $\boldsymbol{x}_k \in \mathcal{K}_k(\boldsymbol{A}, \boldsymbol{r}_0)$ could be written as $\boldsymbol{x}_k = \boldsymbol{K}_k\boldsymbol{c}$, turning the problem into $\min_{\boldsymbol{c}} \|\boldsymbol{b} - \boldsymbol{A}\boldsymbol{K}_k\boldsymbol{c}\|_2$. However, the columns $\boldsymbol{A}^j\boldsymbol{r}_0$ rapidly align with the dominant eigenvector of $\boldsymbol{A}$ (power iteration!), making $\boldsymbol{K}_k$ severely ill-conditioned.

Instead, GMRES builds an **orthonormal basis** $\boldsymbol{V}_k = (\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_k)$ (analogous to the $\boldsymbol{Q}_k$ factor in $\boldsymbol{K}_k = \boldsymbol{Q}_k\boldsymbol{R}_k$) of $\mathcal{K}_k(\boldsymbol{A}, \boldsymbol{r}_0)$ step by step using Gram–Schmidt orthogonalization:

$$\begin{aligned}
\boldsymbol{v}_1 &= \frac{\boldsymbol{r}_0}{\|\boldsymbol{r}_0\|_2}, \\
\boldsymbol{v}_2 &= \frac{\boldsymbol{A}\boldsymbol{v}_1 - ((\boldsymbol{A}\boldsymbol{v}_1)^T\boldsymbol{v}_1)\boldsymbol{v}_1}{\|\boldsymbol{A}\boldsymbol{v}_1 - ((\boldsymbol{A}\boldsymbol{v}_1)^T\boldsymbol{v}_1)\boldsymbol{v}_1\|_2}, \\
\boldsymbol{v}_3 &= \frac{\boldsymbol{A}\boldsymbol{v}_2 - ((\boldsymbol{A}\boldsymbol{v}_2)^T\boldsymbol{v}_1)\boldsymbol{v}_1 - ((\boldsymbol{A}\boldsymbol{v}_2)^T\boldsymbol{v}_2)\boldsymbol{v}_2}{\|\boldsymbol{A}\boldsymbol{v}_2 - ((\boldsymbol{A}\boldsymbol{v}_2)^T\boldsymbol{v}_1)\boldsymbol{v}_1 - ((\boldsymbol{A}\boldsymbol{v}_2)^T\boldsymbol{v}_2)\boldsymbol{v}_2\|_2}, \quad \ldots
\end{aligned}$$

6.3 Arnoldi Orthogonalization and the Hessenberg Decomposition

Formalizing this Gram–Schmidt process on the Krylov subspace yields **Arnoldi's algorithm** (written in Modified Gram–Schmidt form for numerical stability):

Algorithm 1: Arnoldi Orthogonalization

```
1 v_1 = r_0/‖r_0‖_2
2 for j = 1, ..., m do
3 |   w = A v_j
4 |   for k = 1, ..., j do
5 |   |   h_{k,j} = w^T v_k
6 |   |   w = w − h_{k,j} v_k
7 |   h_{j+1,j} = ‖w‖_2
8 |   v_{j+1} = w/h_{j+1,j}
```

From lines 7–8 of the Arnoldi algorithm, at step j we have

$$\boldsymbol{w} = h_{j+1,j}\boldsymbol{v}_{j+1} = \boldsymbol{A}\boldsymbol{v}_j - \sum_{k=1}^{j} h_{k,j}\boldsymbol{v}_k \quad \Longleftrightarrow \quad \boldsymbol{A}\boldsymbol{v}_j = \sum_{k=1}^{j+1} h_{k,j}\boldsymbol{v}_k.$$

Writing this relation in matrix form for $j = 1, \ldots, m$ with $\boldsymbol{V}_m = (\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m) \in \mathbb{R}^{n \times m}$ and $\boldsymbol{V}_{m+1} = (\boldsymbol{v}_1, \ldots, \boldsymbol{v}_{m+1}) \in \mathbb{R}^{n \times (m+1)}$ gives the **Arnoldi relation**:

$$\boldsymbol{A}\boldsymbol{V}_m = (\boldsymbol{A}\boldsymbol{v}_1,\ \boldsymbol{A}\boldsymbol{v}_2,\ \ldots,\ \boldsymbol{A}\boldsymbol{v}_m) = \left(\sum_{k=1}^{2} h_{k,1}\boldsymbol{v}_k,\ \sum_{k=1}^{3} h_{k,2}\boldsymbol{v}_k,\ \ldots,\ \sum_{k=1}^{m+1} h_{k,m}\boldsymbol{v}_k\right) = \boldsymbol{V}_{m+1}\widetilde{\boldsymbol{H}}_m, \tag{21}$$

where $\widetilde{\boldsymbol{H}}_m \in \mathbb{R}^{(m+1)\times m}$ is an **upper Hessenberg matrix**:

$$\widetilde{\boldsymbol{H}}_m = \begin{pmatrix} h_{1,1} & h_{1,2} & \cdots & h_{1,m} \\ h_{2,1} & h_{2,2} & \cdots & h_{2,m} \\ 0 & h_{3,2} & \ddots & \vdots \\ \vdots & \ddots & \ddots & h_{m,m} \\ 0 & \cdots & 0 & h_{m+1,m} \end{pmatrix}.$$

6.4 Reducing GMRES to a Small Least-Squares Problem

At iteration m (or $i+1$), any candidate solution $\boldsymbol{x}_m \in \boldsymbol{x}_0 + \mathcal{K}_m(\boldsymbol{A}, \boldsymbol{r}_0)$ can be written as

$$\boldsymbol{x}_m = \boldsymbol{x}_0 + \boldsymbol{V}_m \boldsymbol{y}, \qquad \boldsymbol{y} \in \mathbb{R}^m.$$

Using the Arnoldi relation $\boldsymbol{A}\boldsymbol{V}_m = \boldsymbol{V}_{m+1}\widetilde{\boldsymbol{H}}_m$ and the fact that $\boldsymbol{r}_0 = \|\boldsymbol{r}_0\|_2 \boldsymbol{v}_1 = \boldsymbol{V}_{m+1}(\|\boldsymbol{r}_0\|_2 \boldsymbol{e}_1)$ (where $\boldsymbol{e}_1 = (1, 0, \ldots, 0)^T \in \mathbb{R}^{m+1}$), the residual norm simplifies dramatically:

$$\begin{aligned} \|\boldsymbol{b} - \boldsymbol{A}\boldsymbol{x}_m\|_2 &= \|\boldsymbol{b} - \boldsymbol{A}(\boldsymbol{x}_0 + \boldsymbol{V}_m \boldsymbol{y})\|_2 = \|\boldsymbol{r}_0 - \boldsymbol{A}\boldsymbol{V}_m \boldsymbol{y}\|_2 \\ &= \|\boldsymbol{r}_0 - \boldsymbol{V}_{m+1}\widetilde{\boldsymbol{H}}_m \boldsymbol{y}\|_2 = \left\| \boldsymbol{V}_{m+1} \left(\|\boldsymbol{r}_0\|_2 \boldsymbol{e}_1 - \widetilde{\boldsymbol{H}}_m \boldsymbol{y} \right) \right\|_2. \end{aligned}$$

Because $\boldsymbol{V}_{m+1}$ has orthonormal columns ($\boldsymbol{V}_{m+1}^T \boldsymbol{V}_{m+1} = \boldsymbol{I}_{m+1}$), it preserves Euclidean norms for any vector $\boldsymbol{z} \in \mathbb{R}^{m+1}$:

$$\|\boldsymbol{V}_{m+1}\boldsymbol{z}\|_2^2 = \boldsymbol{z}^T \boldsymbol{V}_{m+1}^T \boldsymbol{V}_{m+1} \boldsymbol{z} = \boldsymbol{z}^T \boldsymbol{I} \boldsymbol{z} = \|\boldsymbol{z}\|_2^2.$$

Therefore, the n-dimensional minimization problem reduces to a tiny $(m+1) \times m$ least-squares problem:

$$\boxed{\|\boldsymbol{b} - \boldsymbol{A}\boldsymbol{x}_m\|_2 = \left\| \|\boldsymbol{r}_0\|_2 \boldsymbol{e}_1 - \widetilde{\boldsymbol{H}}_m \boldsymbol{y} \right\|_2.} \tag{22}$$

Since $m \ll n$, this small upper Hessenberg least-squares problem $\min_{\boldsymbol{y}\in\mathbb{R}^m} \|\boldsymbol{c} - \widetilde{\boldsymbol{H}}_m \boldsymbol{y}\|_2$ (with $\boldsymbol{c} = \|\boldsymbol{r}_0\|_2 \boldsymbol{e}_1$) is solved cheaply at each iteration using a QR factorization (via Givens rotations) $\boldsymbol{Q}\widetilde{\boldsymbol{H}}_m = \boldsymbol{R}$:

$$\|\boldsymbol{c} - \widetilde{\boldsymbol{H}}_m \boldsymbol{y}\|_2 = \|\boldsymbol{Q}(\boldsymbol{c} - \widetilde{\boldsymbol{H}}_m \boldsymbol{y})\|_2 = \|\boldsymbol{Q}\boldsymbol{c} - \boldsymbol{R}\boldsymbol{y}\|_2,$$

where the last component of $\boldsymbol{Q}\boldsymbol{c} = \|\boldsymbol{r}_0\|_2 \boldsymbol{Q}\boldsymbol{e}_1$ immediately gives the current residual norm $\|\boldsymbol{r}_m\|_2$ without even forming $\boldsymbol{x}_m$ until convergence!